

reference format

**Font format: Arial, Front size:11**

**Date:**

**Title:**

**Purpose:**

**Procedure:**

1. Heat the top of the glass vial with the Bunsen burner.
2. Add sterile water and crack it with tweezers.
3. Remove paper and cotton.
4. Add 500  $\mu$ L LB medium and resuspend the tablet.
5. Drop the supernatant and streak it with four-quadrant streak method.
6. Spread 100  $\mu$ L supernatant on LB plate with 100  $\mu$ L 25 mg/mL chloramphenicol.
7. Inoculate the remaining supernatant in 5 mL LB medium with 25  $\mu$ g/mL chloramphenicol.
8. Incubate in 37°C, started from 18:00

Done by Chars, recorded by Chars.

**Results:**

April



2026/4/22



**Date:** 20260422

**Title:** Recover *E. coli* harboring pNW33N from BCRC

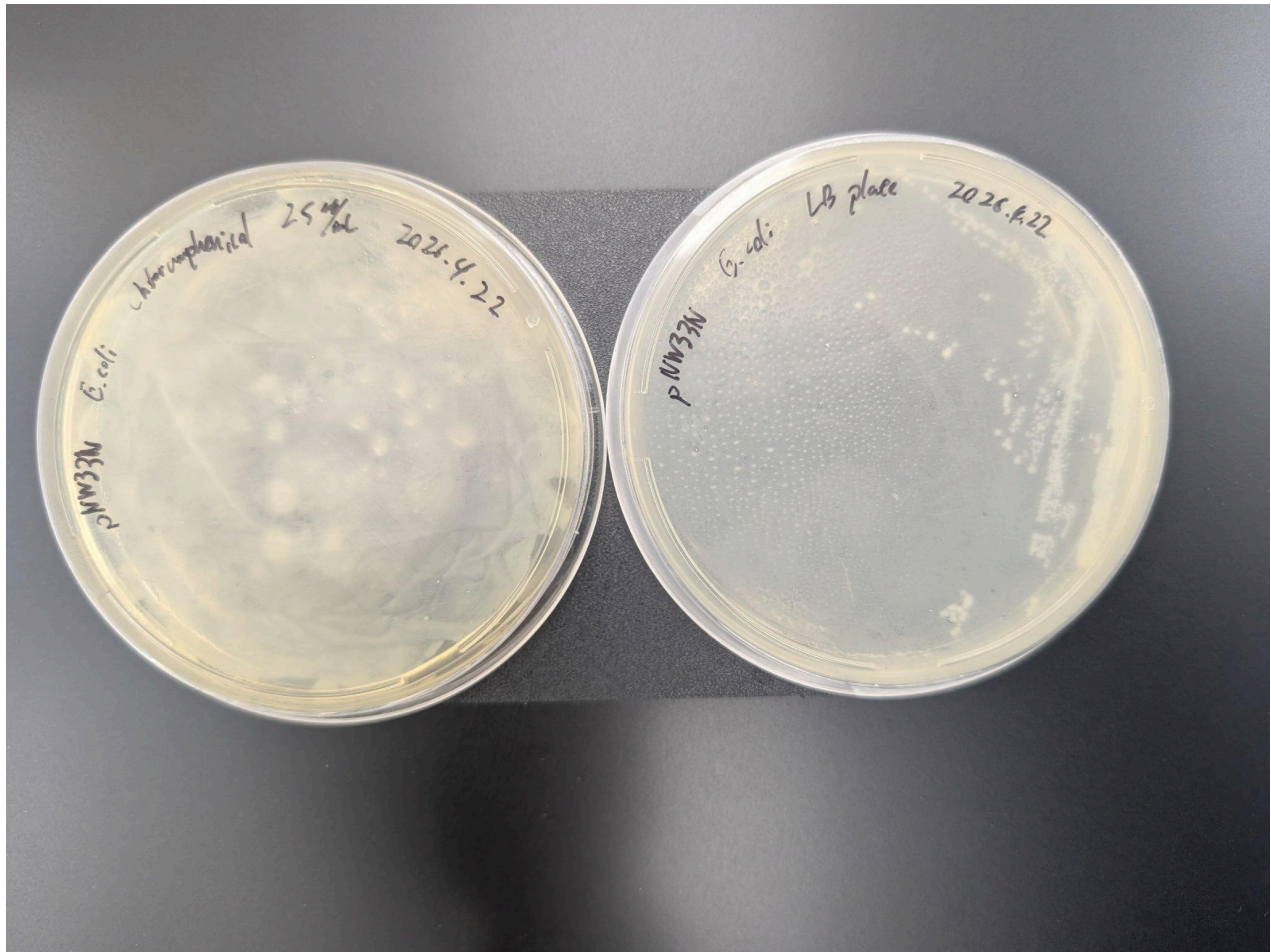
**Purpose:** Recover *E. coli* harboring the shuttle vector pNW33N, which was obtained from BCRC.

**Procedure:**

1. Heat the top of the glass vial with the Bunsen burner.
2. Add sterile water and crack it with tweezers.
3. Remove paper and cotton.
4. Add 500  $\mu$ L LB medium and resuspend the tablet.
5. Drop the supernatant and streak it with four-quadrant streak method.
6. Spread 100  $\mu$ L supernatant on LB plate with 100  $\mu$ L 25 mg/mL chloramphenicol.
7. Inoculate the remaining supernatant in 5 mL LB medium with 25  $\mu$ g/mL chloramphenicol.
8. Incubate in 37°C, started from 18:00

Done by Chars, recorded by Chars.

**Results:**



Recorded by Chars

2026/4/23



**Date: 20260423**

**Title: Miniprep**

**Purpose: Using miniprep kit (Viogene) to isolate pNW33N plasmid from *E. coli***

**Procedure:**

1. Collected cells from liquid culture by centrifuge 23444 g for 1 min.
2. Repeat the previous step until all of the liquid cultures were collected.
3. Add 200  $\mu$ L MX1 buffer and resuspend pellet by pipetting.
4. Add 250  $\mu$ L MX2 buffer and gently invert 8 times.
5. Incubated at room temperature for 2 min.
6. Add 350  $\mu$ L MX3 buffer and gently invert 8 times.
7. Centrifuge at 23444 g for 5 min.
8. Transfer the supernatant to the column by P200 micropipette.
9. Centrifuge at 23444 g for 30 sec.
10. Discard the flow-through. Add 500  $\mu$ L WN buffer.
11. Centrifuge at 23444 g for 30 sec.
12. Discard the flow-through. Add 700  $\mu$ L WS buffer.
13. Centrifuge at 23444 g for 30 sec.
14. Discard the flow-through. Centrifuge the empty column at 23444 g for 3 min.
15. Add 50  $\mu$ L EB buffer at 65°C. Incubated at room temperature for 5 min.
16. Centrifuge at 23444 g for 3 min.
17. Measure the concentration by Biodrop

**Results:**

| Plasmid   | Concentration (ng/ $\mu$ L) | A260/280 | A260/230 |
|-----------|-----------------------------|----------|----------|
| pNW33N #1 | 71                          | 1.775    | 0.910    |
| pNW33N #2 | 78                          | 1.660    | 0.684    |

Done by Chars, recorded by Chars.

**Date: 20260423**

**Title: Enzyme Digestion**

**Purpose:**

**(1) Verify pNW33N obtained from BCRC**

**(2) Prepare vector for Gibson assembly**

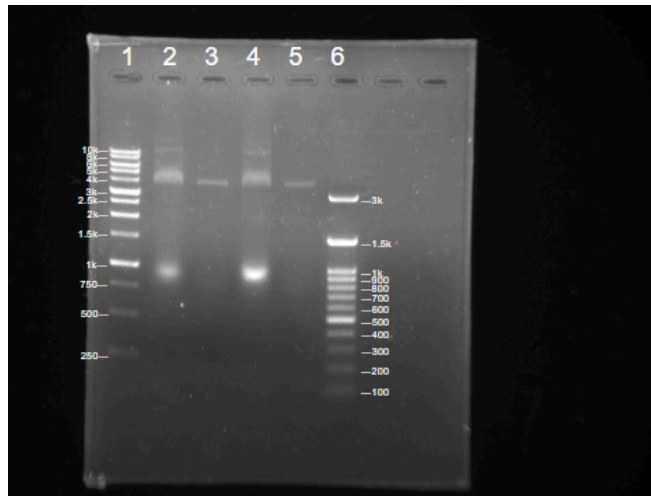
**Procedure:**

1. Prepare the reaction.

|                     | #1 (μL) | #2 (μL) |
|---------------------|---------|---------|
| Vector (1 μg)       | 14.1    | 12.8    |
| 10x CutSmart buffer | 5       | 5       |
| Distilled water     | 28.9    | 30.2    |
| EcoRI-HF            | 1       | 1       |
| BtsI-v2             | 1       | 1       |

2. Incubate in a PCR machine at 37°C for 2 hours.
3. Resolve in 1% agarose gel in 1x TAE for 30 min.

**Results:**



- 1: Omics 1kb DNA RTU Ladder 5 uL
- 2: pNW33N #1 5 uL (355 ng)
- 3: pNW33N #1 EcoRI BtsI 2hr 5 uL (100 ng)
- 4: pNW33N #2 5 uL (390 ng)
- 5: pNW33N #2 EcoRI BtsI 2hr 5 uL (100 ng)
- 6: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL

Done by Chars, recorded by Chars

2026/4/27



**Date: 20260427**

**Title: Hydroponic System Testing**

**Purpose: Prepare hydroponic system for testing**

**Preparation of 40 ml ½ MS agar**

**Procedure:**

1. 1/2MS agar preparation
  - a. 1.58 g MS powder \_ 40 mL ddH<sub>2</sub>O
  - b. Sterilize the mixture using autoclave
2. Prepare the holder (1.5ml tube lid)
  - a. Use a leather punch to make 3-mm holes in the lid.
  - b. Pour the agar into 1.5 ml eppendorf's lids and wait for solidification.
3. Sow one seed in each hole of the lid. Vernalization
  - a. Put the seed-containing holder at 4°C for 3 days (wrap with foil)

Done by Gabriel, Recorded by Sophie L

**Date: 20260427**

**Title: Preparation of PCR-Amplified sfGFP fragments**

**Purpose: linear DNA template for quick cell-free expression**

**Procedure:**

1. Prepare the reaction as follows:

|                                 |         |
|---------------------------------|---------|
| 5x Q5 Reaction buffer           | 10 µL   |
| 2.5 mM dNTP mix                 | 4 µL    |
| 10 µM pNW33N-L2(sfGFP) F primer | 2.5 µL  |
| 10 µM pNW33N-L2(sfGFP) R primer | 2.5 µL  |
| 70 µg/mL pJUMP49-2B plasmid     | 1 µL    |
| Q5 Hi-Fi DNA polymerase         | 0.5 µL  |
| ddH <sub>2</sub> O              | 29.5 µL |
| Total                           | 50 µL   |

2. PCR program

|                          |      |        |
|--------------------------|------|--------|
| Denaturation             | 98°C | 30 sec |
| Denaturation (25 cycles) | 98°C | 10 sec |
| Annealing (25 cycles)    | 60°C | 30 sec |

|                       |      |       |
|-----------------------|------|-------|
| Extension (25 cycles) | 72°C | 1 min |
| Extension             | 72°C | 2 min |

3. Storage PCR product at 4°C
4. PCR product were confirmed by gel electrophoresis (Amplicon size:1211 bp)

Done by Gabriel, Recorded by Sophie

2026/4/28 

**Date: 20260428**

**Title: Cleanup of PCR or Restriction enzyme digestion products Purpose: Remove contaminants** Mix the PCR product (or enzyme digestion product) with 0.5 mL GP buffer

1. Transfer the mixture into GP column
2. Centrifuge 1 min at 7000g, then discard the flow-through
3. Add 0.5 mL WN buffer into the column
4. Centrifuge 1 min at 7000g, then discard the flow-through
5. Add 0.5 mL WS buffer into the column
6. Centrifuge 1 min at 7000g, then discard the flow-through
7. Centrifuge the column at 23444g for 3 min
8. Place the column onto a 1.5 mL centrifuge tube
9. Add 30  $\mu$ L elution buffer onto the membrane and wait for 3 min
10. Centrifuge at 23444g for 2 min
11. Storage at -20°C

Done by Gabriel, Recorded by Alex

**Date: 20260428**

**Title: Gibson assembly and Transformation**

**Purpose: To create the shuttle vector with MoClo 2.0 system**

**Procedure:**

1. Prepare the reaction

| Component                                                 | 20 $\mu$ L reaction | Length (base pair) |
|-----------------------------------------------------------|---------------------|--------------------|
| pNW33N enzyme digestion product(cleaning up) 3 $\mu$ g/mL | 8.26 $\mu$ L        | 4021bp             |
| pJCMP49-2B PCR product (cleaning up) 60 $\mu$ g/mL        | 0.24 $\mu$ L        | 1207bp             |
| NEBuilder HiFi DNA assembly master mix                    | 10 $\mu$ L          |                    |
| ddH <sub>2</sub> O                                        | 1.5 $\mu$ L         |                    |
| total                                                     | 20 $\mu$ L          |                    |

2. Incubate the reaction in PCR machine at 50°C for 15 min
3. Transfer 10 $\mu$ L of the product into 100 $\mu$ L competent cells.
4. On ice for 20 min
5. Heat shock at 42°C for 60sec
6. Put the competent cell/DNA mixture on ice for 2 min
7. Add 1mL SOC medium and culture at 37°C for 1 hr
8. Transfer 100 $\mu$ L culture medium on the LB agar plate with chloramphenicol
9. Spread the transformants on agar plate The plate incubated at 37°C overnight

Done by Gabriel



Date: 20260428

Title: Gibson assembly and transformation

Purpose: Create the shuttle vector with MoClo 2.0 system

Procedure:

1. Prepare reaction

| substance                                              | amount            |
|--------------------------------------------------------|-------------------|
| pNW33N enzyme digestion product (cleaning up) (3µg/mL) | 8.26 µL (4021 bp) |
| pJUMP49-2B PCR product (cleaning up) (60µg/mL)         | 0.24 µL (1207 bp) |
| NEBuiider HiFi DNA assembly master mix                 | 10 µL             |
| ddH2O                                                  | 1.5 µL            |
| total                                                  | 20 µL             |

2. Incubate the reaction in PCR machine at 50°C for 15 min
3. Transfer 10 µL of the product into 100 µL competent cells
4. Stand on ice for 20 min
5. Heat shock at 42°C for 60 sec
6. Put the reaction on ice for 2 min
7. Add 1 mL SOC medium and culture at 37°C for 1hr
8. Transfer 100 µL culture medium on the LB agar plate with chloramphenicol
9. Spread the liquid
10. The plate were incubated at 37°C overnight

Results:

Done by Gabriel, Recorded by Alex

2026/4/29



**Date:20260429**

**Title: Recover Bacillus Subtilis 168 from BCRC**

**Purpose: Recover Bacillus Subtilis 168 from BCRC**

**Procedure:**

1. Heat the top of glass vial with bunsen burner
2. Add sterile water and crack it with tweezer
3. Remove paper and cotton
4. Add 500 $\mu$ L LB medium and resuspend the tablet
5. Spread the supernatant with four-quadrant streak method
6. Inoculate the remaining supernatant in LB medium 5mL
7. Incubate at 37°C, started from 21:30

Done by Chars, recorded by Alex

2026/4/30 

**Date: 20260430**

**Title: Miniprep**

**Purpose: Isolate pNW33N - L2(sfGFP) - v1 C#1 & G#2 from DH5a**

**Procedure:**

1. Collected cells from liquid culture by centrifuge 23444g 1min
2. Repeat the previous step until all liquid culture are collected
3. Add 200 $\mu$ L MX1 buffer and resuspend pellet by pipetting
4. Add 250 $\mu$ L MX2 buffer and gently invert 10 times
5. Incubate at room temperature for 2 min
6. Add 350 MX3 buffer and gently invert 10 times
7. Centrifuge 23444g for 5 min
8. Transfer supernatant to the column by P200
9. Centrifuge 30 sec at 7000g, then discard the flow-through
10. Add 0.5 mL WN buffer into the column
11. Centrifuge 30 sec at 7000g, then discard the flow-through
12. Add 0.7 mL WS buffer into the column
13. Centrifuge 30 sec at 7000g, then discard the flow-through
14. Centrifuge empty column at 10000g for 3 min
15. Add 50  $\mu$ L of EB buffer preheated to 65°C. Incubate at room temperature for 5 min.  
Centrifuge at 10000g for 3min
16. Measure the concentration by Biodrop

**Result:**

| Plasmid               | Concentration (ng/ $\mu$ L) | A260/280 | A260/230 |
|-----------------------|-----------------------------|----------|----------|
| pNW33N - L2(sfGFP) #1 | 41                          | 2.050    | 1.708    |
| pNW33N - L2(sfGFP) #2 | 43                          | 1.955    | 1.720    |

Done by Chars, Recorded by Alex

